1. Project Description

Out project is a web application that analyzes job growth data for counties in Pennsylvania. Users will select a job industry and the application will list the top five counties with the highest predicted job growth rates and will display them as bar graphs.

2. Project Significance

This project is significant because it provides a simplified way to see which counties are most likely to offer a good future for your job industry and have jobs available for the user. Instead of a user analysing a large data set, they can use this web application to predict the future job growth. This makes the tool useful for job seekers, researchers, or anyone interested in the potential job growth.

3. Data Collection

The dataset used in this project comes from the Quarterly Census of Employment and Wages, provided by the U.S. Bureau of Labor Statistics. The data is stored in an Excel file where each sheet represents a different county in Pennsylvania.

Each sheet contains industry-level data, including number of establishments, employment levels, average wages and employment growth percentages.

The application loads all sheets from the Excel file and combines them into a single dataset for processing, while excluding statewide summary data to avoid skewing results.

4. Processing & Feature Engineering

The raw dataset required several preprocessing steps before it could be used for modeling. First, all sheets were read and combined into a single dataset, with a new column added to indicate the county associated with each row, while the statewide sheet was removed to prevent it from distorting the results.

Next, the data was cleaned by converting formatted text values into numeric types. This included removing commas from numbers and converting percentage strings into numerical values. Rows with missing or invalid data were removed to ensure consistency.

Feature engineering focused on three primary variables: employment, wage, and employment growth, which were selected because they represent job availability, job quality, and future opportunity, respectively. Each feature was normalized using Min-Max scaling to ensure comparability.

An Opportunity Index was then computed as a weighted combination of these normalized features. This index serves as the target variable for the machine learning model and represents the overall strength of job opportunities in a given county for a specific industry.

5. Model Development

The model developed in this project is a Random Forest regression model. The input features include normalized employment, wage, and employment growth values, while the output is the predicted Opportunity Index. The model learns the relationship between these features and the computed index.

Random Forest was selected because it is robust, handles nonlinear relationships well, and performs effectively on the kind of data available. The model was trained using the available dataset, with a configuration of 200 decision trees and a fixed random state for reproducibility.

The model's predictions were used to rank counties based on their Opportunity Index scores. There was a bit of issue with the statewide data skewing results at first, but after removing it, the predicted scores ranged more realistically and allowed meaningful comparisons between counties.

The results showed that employment levels had the strongest influence on the final score, followed by growth rates and wages. This aligns with expectations, as job availability is typically the most critical factor in employment decisions.

6. Showcase

Not sure what exactly is desired here for the showcase, although based on the instructions in the assignment I believe its the video showing its usage and functionalities, which is attached in the GitHub.

7. Discussion and Conclusions

The project successfully demonstrates how data analysis and machine learning can be used to extract meaningful insights from economic data.

The preprocessing and feature engineering produce a useful and simple ranking system for identifying which counties are going to have the best potential for each industry. The web interface allows this processed data to be displayed in an easy to understand and use way.

Some limitations with the current model is that it does not take into account external factors such as population trends or cost of living in such areas and thus a user would have to do their own research into those factors when it comes to things such as job searching. Some potential future improvements for the project are:

- Adding more features for better predictions
- Expanding beyond Pennsylvania to other states and their counties

8. AI Disclosure

ChatGPT was used as a supportive tool during development for clarification, debugging assistance, and exploration of algorithmic ideas.

Usage of it consisted of asking it for potential solutions about how to accomplish a given idea when stuck and unable to think of how to start solving a problem (making sure to request pseudo code or text only responses instead of direct code), and prevalently throwing any feedback errors received into it to get a summary of what it actually meant and suggestions for how to solve it.

However, the overall design of the system, the implementation decisions, the testing process, and the actual writing of code was done and validated through independent work, specifically making not to have the AI directly generate code instead of just supporting.

9. Report Quality

The project reflects a thorough application of machine learning principles to a real-world dataset. Each stage of the process, from data collection to model deployment, was carefully implemented and documented.

The report clearly explains the objectives, methodology, and results of the project. It provides sufficient detail to understand how the model was developed and how it functions within the application.

The overall quality of the work demonstrates a strong understanding of both the technical and practical aspects of machine learning. The project is well-structured, logically organized, and effectively communicates its findings.